Nugget Line — process mapping report

C.H. Guenther & Son — Duncanville, TX · 11/13/2026 · Current state

Values shown as NYC are Not Yet Collected. Each result is labeled Verified inputs, Estimate or Incomplete according to the status of its inputs. Estimated savings are not validated until Finance confirms them against actual cost results.

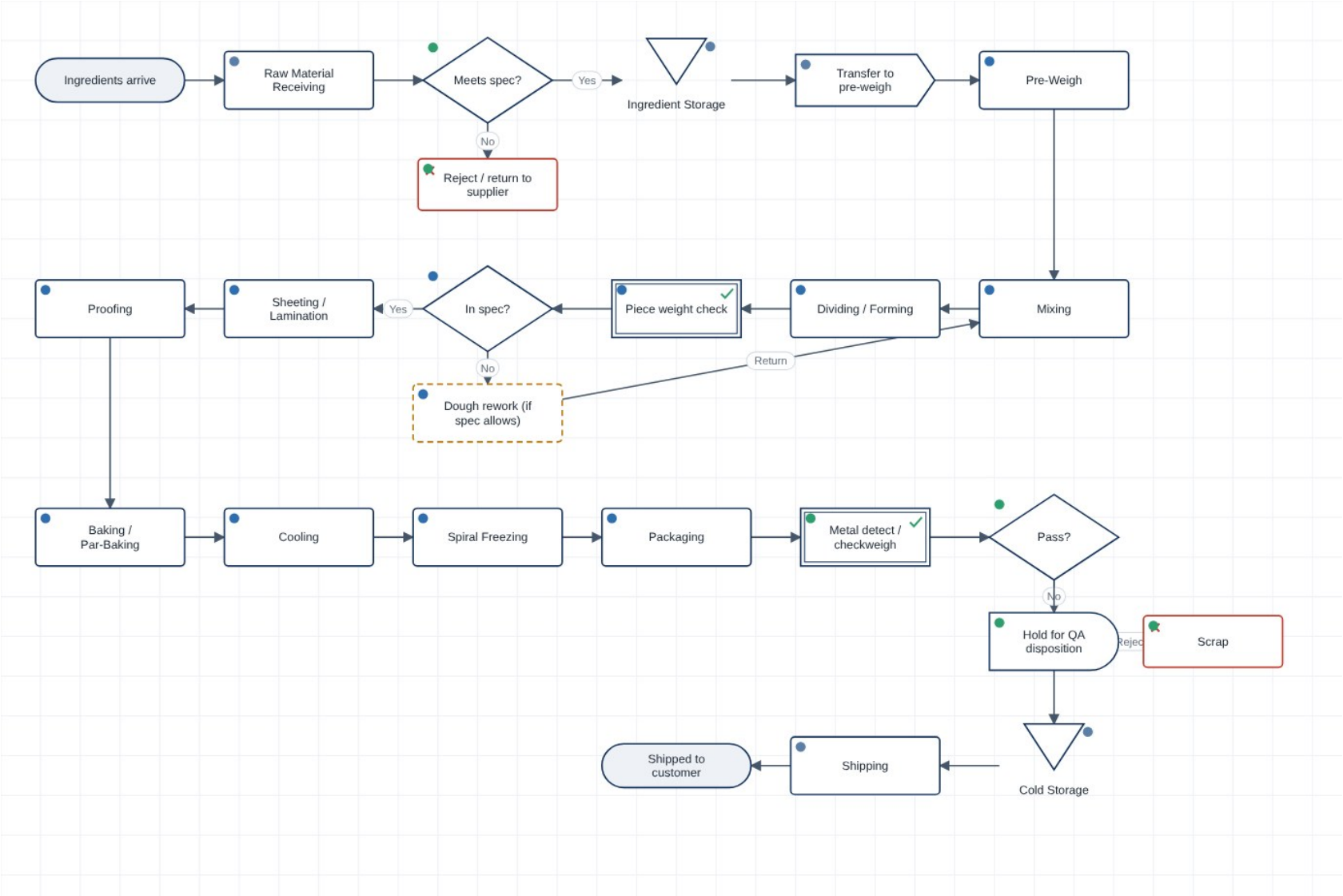
Key performance indicators

Calculated values with data status and missing inputs

KPI	Value	Data status	Missing inputs	Formula
Takt time	NYC	Incomplete	Shifts per day; Shift length; Planned breaks & meetings; Customer demand	Available time per day × 60 ÷ Customer demand per day
Total lead time	NYC	Incomplete	Raw Material Receiving: queue; Raw Material Receiving: processing time; Ingredient Storage: queue; Ingredient Storage: processing time	£ Queue time + £ Processing time over applicable step
Value-added time	NYC	Incomplete	Mixing; Dividing / Forming; Sheeting / Lamination; Proofing	£ Processing time of steps classified Value-added
Process cycle efficiency	NYC	Incomplete	Raw Material Receiving: queue; Raw Material Receiving: processing time; Ingredient Storage: queue; Ingredient Storage: processing time	Value-added time ÷ Total lead time
Bottleneck effective C/T	NYC	Incomplete	Cycle time at every step	Step with the highest effective cycle time among steps that apply to the line
Line capacity	NYC	Incomplete	Cycle time at every step	3600 ÷ Bottleneck effective cycle time
Rolled throughput yield	NYC	Incomplete	Raw Material Receiving; Ingredient Storage; Pre-Weigh; Mixing	(Yield) over steps; where yield is missing, (1" Scrap
Changeover time	NYC	Incomplete	Changeover time and changeovers/week	£ (Changeover time × Changeovers per week) ÷ 60
Travel labor	NYC	Incomplete		Travel time per shift × People × Shifts/day × Operating days/year ÷ 60
Travel labor cost	NYC	Incomplete		Annual travel labor hours × Loaded labor rate

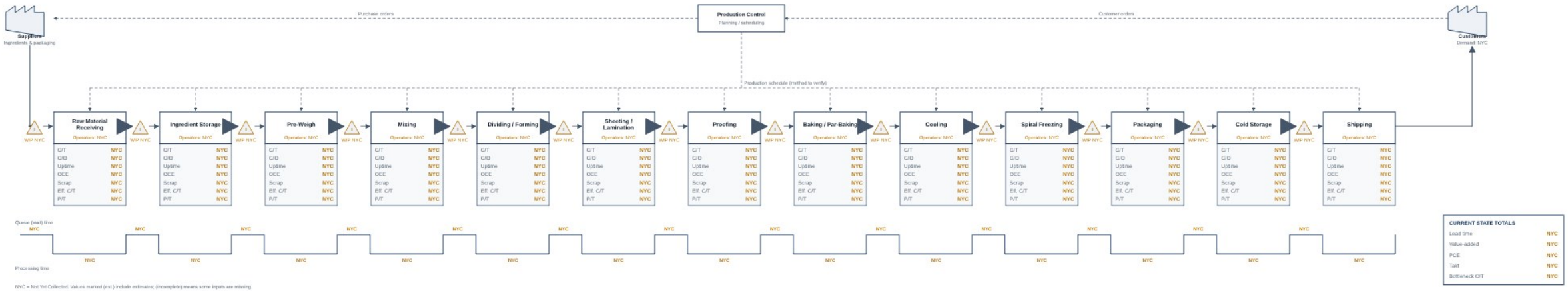
Process flow map

Operations, inspections, decisions, rework and scrap.



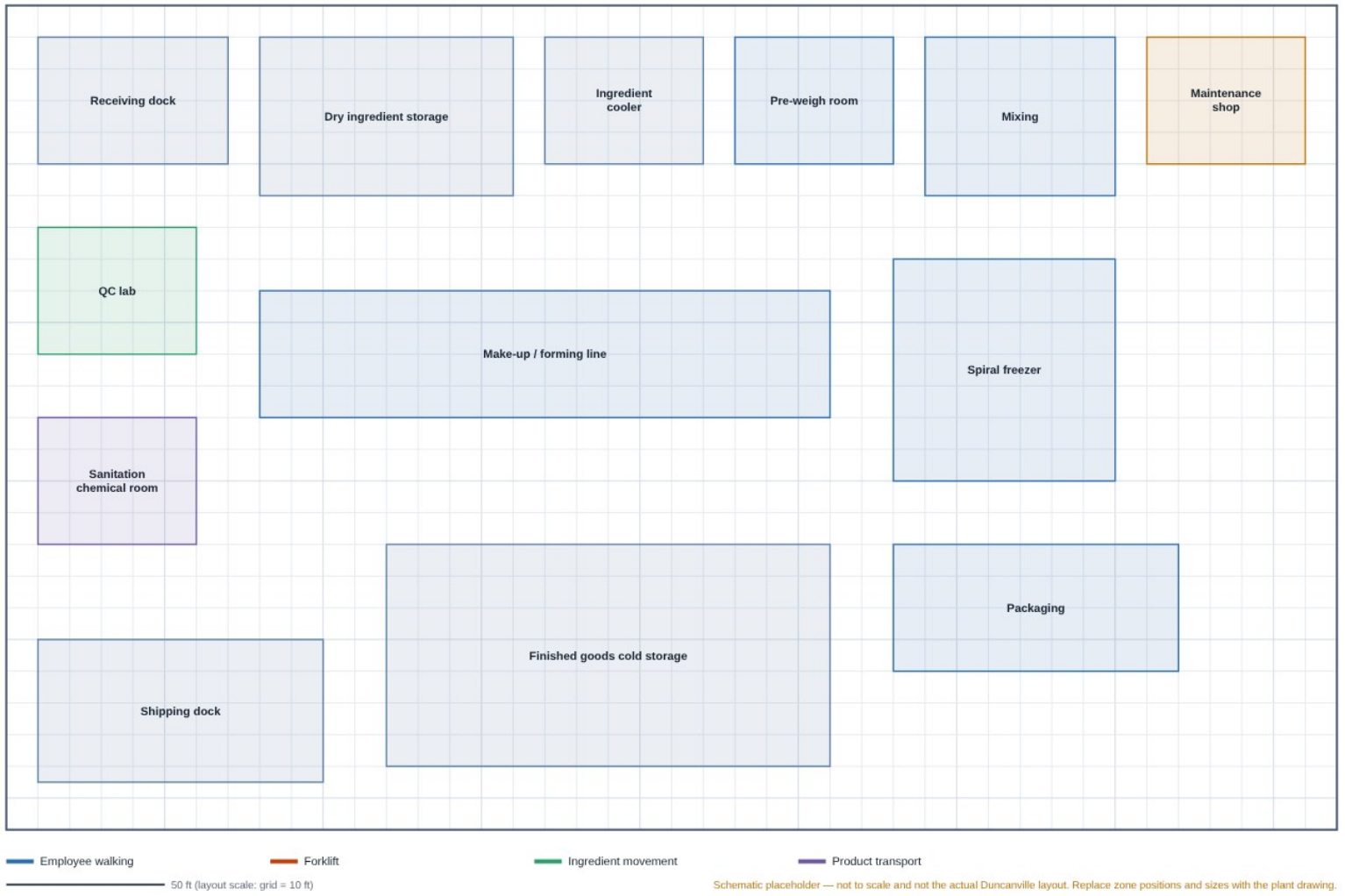
Value stream map

Material and information flow with data boxes and lead-time ladder.



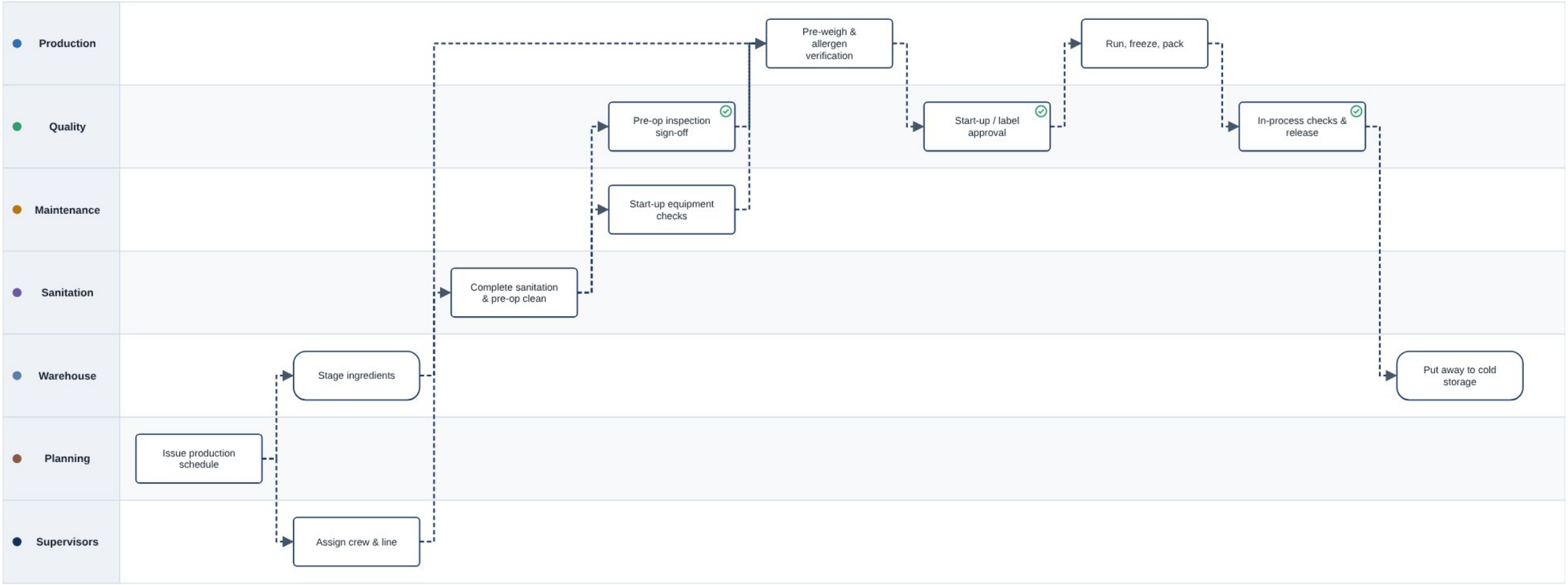
Spaghetti diagram

Schematic placeholder — not to scale and not the actual Duncanville layout. Replace zone positions and sizes with the plant drawing.



Swimlane process map

Responsibilities, handoffs and flagged approvals or delays.



Cross-department handoffs: 12 · Approvals: 3 · Flagged items: 0 · DUP = duplicated work, APPR = unnecessary approval, WAIT = delay, GAP = communication gap

Step data

NYC = Not Yet Collected · (E) estimate · (V) verified

Step	Applies	Cycle time (s/unit)	Processing time (value-added) (min)	Changeover time (min)	Uptime (%)	Labor (people)	Throughput (case/hr)	Scrap (%)	WIP before step (case)	OEE (%)
Raw Material Receiving	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Ingredient Storage	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Pre-Weigh	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Mixing	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Dividing / Forming	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Sheeting / Lamination	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Proofing	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Baking / Par-Baking	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Cooling	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Spiral Freezing	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Packaging	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Cold Storage	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Shipping	To verify	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC

Process activity map

Total 0 min · NVA 0 min · 0 ft · 50 activities not timed

Step	Activity	Type	Time (min)	Distance (ft)	People	Class	Data
Raw Material Receiving	Wait / queue before raw material receiving	Delay	NYC			NVA	Not Yet Collected
Raw Material Receiving	Raw Material Receiving	Operation	NYC			NNVA	Not Yet Collected
Raw Material Receiving	Quality / food-safety check at raw material receiving	Inspection	NYC			NNVA	Not Yet Collected
Ingredient Storage	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Ingredient Storage	Wait / queue before ingredient storage	Delay	NYC			NVA	Not Yet Collected
Ingredient Storage	Held in ingredient storage	Storage	NYC			NVA	Not Yet Collected
Ingredient Storage	Quality / food-safety check at ingredient storage	Inspection	NYC			NNVA	Not Yet Collected
Pre-Weigh	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Pre-Weigh	Wait / queue before pre-weigh	Delay	NYC			NVA	Not Yet Collected
Pre-Weigh	Pre-Weigh	Operation	NYC			NNVA	Not Yet Collected
Pre-Weigh	Quality / food-safety check at pre-weigh	Inspection	NYC			NNVA	Not Yet Collected
Mixing	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Mixing	Wait / queue before mixing	Delay	NYC			NVA	Not Yet Collected
Mixing	Mixing	Operation	NYC			VA	Not Yet Collected
Mixing	Quality / food-safety check at mixing	Inspection	NYC			NNVA	Not Yet Collected
Dividing / Forming	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Dividing / Forming	Wait / queue before dividing / forming	Delay	NYC			NVA	Not Yet Collected
Dividing / Forming	Dividing / Forming	Operation	NYC			VA	Not Yet Collected
Dividing / Forming	Quality / food-safety check at dividing / forming	Inspection	NYC			NNVA	Not Yet Collected
Sheeting / Lamination	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Sheeting / Lamination	Wait / queue before sheeting / lamination	Delay	NYC			NVA	Not Yet Collected
Sheeting / Lamination	Sheeting / Lamination	Operation	NYC			VA	Not Yet Collected
Proofing	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Proofing	Wait / queue before proofing	Delay	NYC			NVA	Not Yet Collected
Proofing	Proofing	Operation	NYC			VA	Not Yet Collected
Proofing	Quality / food-safety check at proofing	Inspection	NYC			NNVA	Not Yet Collected

Step	Activity	Type	Time (min)	Distance (ft)	People	Class	Data
Baking / Par-Baking	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Baking / Par-Baking	Wait / queue before baking / par-baking	Delay	NYC			NVA	Not Yet Collected
Baking / Par-Baking	Baking / Par-Baking	Operation	NYC			VA	Not Yet Collected
Baking / Par-Baking	Quality / food-safety check at baking / par-baking	Inspection	NYC			NNVA	Not Yet Collected
Cooling	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Cooling	Wait / queue before cooling	Delay	NYC			NVA	Not Yet Collected
Cooling	Cooling	Operation	NYC			NNVA	Not Yet Collected
Cooling	Quality / food-safety check at cooling	Inspection	NYC			NNVA	Not Yet Collected
Spiral Freezing	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Spiral Freezing	Wait / queue before spiral freezing	Delay	NYC			NVA	Not Yet Collected
Spiral Freezing	Spiral Freezing	Operation	NYC			VA	Not Yet Collected
Spiral Freezing	Quality / food-safety check at spiral freezing	Inspection	NYC			NNVA	Not Yet Collected
Packaging	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Packaging	Wait / queue before packaging	Delay	NYC			NVA	Not Yet Collected
Packaging	Packaging	Operation	NYC			VA	Not Yet Collected
Packaging	Quality / food-safety check at packaging	Inspection	NYC			NNVA	Not Yet Collected
Cold Storage	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Cold Storage	Wait / queue before cold storage	Delay	NYC			NVA	Not Yet Collected
Cold Storage	Held in cold storage	Storage	NYC			NVA	Not Yet Collected
Cold Storage	Quality / food-safety check at cold storage	Inspection	NYC			NNVA	Not Yet Collected
Shipping	Move from previous step	Transportation	NYC			NVA	Not Yet Collected
Shipping	Wait / queue before shipping	Delay	NYC			NVA	Not Yet Collected
Shipping	Shipping	Operation	NYC			NNVA	Not Yet Collected
Shipping	Quality / food-safety check at shipping	Inspection	NYC			NNVA	Not Yet Collected

Waste & opportunity register, actions, finance validation

Estimated savings are not validated until Finance confirms them.

Step	Waste	Opportunity	Est. $/yr	Benefit type	Finance status	Validated $/yr

Floor-walk checklist — open items

256 items still to collect

Area	Item	How to measure	Done
Line settings (Planning / Finance)	Customer demand	Planning: average daily requirement for this line, last 13 weeks.	
Line settings (Planning / Finance)	Shifts per day	Planning / production schedule.	
Line settings (Planning / Finance)	Shift length	Scheduled minutes per shift.	
Line settings (Planning / Finance)	Planned breaks & meetings	Scheduled breaks, lunch, huddles, planned sanitation inside the shift.	
Line settings (Planning / Finance)	Operating days per year	Planning: production calendar.	
Line settings (Planning / Finance)	Loaded labor rate	Finance only. Do not use an estimate for reported savings.	
Line settings (Planning / Finance)	Material cost per unit	Finance: standard material cost per unit (for scrap valuation at cost).	
Line settings (Planning / Finance)	Forklift / pallet-jack speed	Observe loaded travel; plant speed limits apply.	
Raw Material Receiving	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Raw Material Receiving	Process owner	Confirm with the area supervisor.	
Raw Material Receiving	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Raw Material Receiving	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Raw Material Receiving	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Raw Material Receiving	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Raw Material Receiving	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Raw Material Receiving	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Raw Material Receiving	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Raw Material Receiving	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Raw Material Receiving	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	

Area	Item	How to measure	Done
Raw Material Receiving	Yield (%)	Good output ÷ input (weight basis), from production records.	
Raw Material Receiving	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Raw Material Receiving	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Raw Material Receiving	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Raw Material Receiving	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Raw Material Receiving	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Raw Material Receiving	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Raw Material Receiving	Verify food-safety controls, allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Ingredient Storage	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Ingredient Storage	Process owner	Confirm with the area supervisor.	
Ingredient Storage	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Ingredient Storage	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Ingredient Storage	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
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Ingredient Storage	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
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Ingredient Storage	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Ingredient Storage	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Ingredient Storage	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Ingredient Storage	Verify food-safety controls, allergen controls, traceability records, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	

Area	Item	How to measure	Done
Pre-Weigh	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Pre-Weigh	Process owner	Confirm with the area supervisor.	
Pre-Weigh	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Pre-Weigh	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Pre-Weigh	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Pre-Weigh	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Pre-Weigh	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Pre-Weigh	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Pre-Weigh	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Pre-Weigh	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Pre-Weigh	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Pre-Weigh	Yield (%)	Good output ÷ input (weight basis), from production records.	
Pre-Weigh	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Pre-Weigh	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Pre-Weigh	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Pre-Weigh	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Pre-Weigh	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Pre-Weigh	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Pre-Weigh	Verify allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Mixing	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Mixing	Process owner	Confirm with the area supervisor.	
Mixing	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Mixing	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Mixing	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Mixing	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Mixing	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	

Area	Item	How to measure	Done
Mixing	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Mixing	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Mixing	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Mixing	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Mixing	Yield (%)	Good output ÷ input (weight basis), from production records.	
Mixing	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Mixing	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Mixing	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Mixing	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Mixing	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Mixing	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Mixing	Verify traceability records, quality checks, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Dividing / Forming	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Dividing / Forming	Process owner	Confirm with the area supervisor.	
Dividing / Forming	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Dividing / Forming	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Dividing / Forming	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Dividing / Forming	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Dividing / Forming	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Dividing / Forming	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Dividing / Forming	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Dividing / Forming	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Dividing / Forming	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Dividing / Forming	Yield (%)	Good output ÷ input (weight basis), from production records.	
Dividing / Forming	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Dividing / Forming	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Dividing / Forming	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	

Area	Item	How to measure	Done
Dividing / Forming	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Dividing / Forming	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Dividing / Forming	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Dividing / Forming	Verify food-safety controls, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Sheeting / Lamination	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Sheeting / Lamination	Process owner	Confirm with the area supervisor.	
Sheeting / Lamination	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Sheeting / Lamination	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Sheeting / Lamination	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Sheeting / Lamination	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Sheeting / Lamination	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Sheeting / Lamination	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Sheeting / Lamination	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Sheeting / Lamination	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Sheeting / Lamination	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Sheeting / Lamination	Yield (%)	Good output ÷ input (weight basis), from production records.	
Sheeting / Lamination	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Sheeting / Lamination	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Sheeting / Lamination	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Sheeting / Lamination	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Sheeting / Lamination	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Sheeting / Lamination	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Proofing	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Proofing	Process owner	Confirm with the area supervisor.	
Proofing	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Proofing	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	

Area	Item	How to measure	Done
Proofing	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Proofing	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Proofing	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Proofing	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Proofing	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Proofing	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Proofing	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Proofing	Yield (%)	Good output ÷ input (weight basis), from production records.	
Proofing	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Proofing	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Proofing	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Proofing	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Proofing	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Proofing	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Proofing	Verify quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Baking / Par-Baking	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Baking / Par-Baking	Process owner	Confirm with the area supervisor.	
Baking / Par-Baking	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Baking / Par-Baking	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Baking / Par-Baking	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Baking / Par-Baking	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Baking / Par-Baking	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Baking / Par-Baking	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Baking / Par-Baking	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Baking / Par-Baking	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Baking / Par-Baking	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Baking / Par-Baking	Yield (%)	Good output ÷ input (weight basis), from production records.	

Area	Item	How to measure	Done
Baking / Par-Baking	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Baking / Par-Baking	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Baking / Par-Baking	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Baking / Par-Baking	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Baking / Par-Baking	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Baking / Par-Baking	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Baking / Par-Baking	Verify food-safety controls	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Cooling	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Cooling	Process owner	Confirm with the area supervisor.	
Cooling	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Cooling	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Cooling	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Cooling	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Cooling	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Cooling	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Cooling	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Cooling	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Cooling	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Cooling	Yield (%)	Good output ÷ input (weight basis), from production records.	
Cooling	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Cooling	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Cooling	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Cooling	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Cooling	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Cooling	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Cooling	Verify food-safety controls	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Spiral Freezing	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	

Area	Item	How to measure	Done
Spiral Freezing	Process owner	Confirm with the area supervisor.	
Spiral Freezing	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Spiral Freezing	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Spiral Freezing	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Spiral Freezing	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Spiral Freezing	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Spiral Freezing	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Spiral Freezing	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Spiral Freezing	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Spiral Freezing	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Spiral Freezing	Yield (%)	Good output ÷ input (weight basis), from production records.	
Spiral Freezing	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Spiral Freezing	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Spiral Freezing	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Spiral Freezing	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Spiral Freezing	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Spiral Freezing	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Spiral Freezing	Verify quality checks, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Packaging	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Packaging	Process owner	Confirm with the area supervisor.	
Packaging	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Packaging	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Packaging	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Packaging	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Packaging	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Packaging	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.

Area	Item	How to measure	Done
Packaging	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Packaging	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Packaging	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Packaging	Yield (%)	Good output ÷ input (weight basis), from production records.	
Packaging	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Packaging	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Packaging	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Packaging	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Packaging	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Packaging	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Packaging	Verify food-safety controls, allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Cold Storage	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Cold Storage	Process owner	Confirm with the area supervisor.	
Cold Storage	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Cold Storage	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Cold Storage	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Cold Storage	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Cold Storage	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Cold Storage	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Cold Storage	Labor (people)	Operators assigned to the step per shift (staffing plan), confirmed by observation.	
Cold Storage	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Cold Storage	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Cold Storage	Yield (%)	Good output ÷ input (weight basis), from production records.	
Cold Storage	WIP before step (cases)	Physical count of product or batches waiting in front of the step at the time of the walk; repeat at 2–3 times.	
Cold Storage	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Cold Storage	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	

Area	Item	How to measure	Done
Cold Storage	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Cold Storage	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Cold Storage	Verify food-safety controls, traceability records	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Shipping	Confirm whether this step exists on the line, and its equipment	Walk the line with the supervisor; record equipment name and asset number.	
Shipping	Process owner	Confirm with the area supervisor.	
Shipping	Cycle time (s/unit)	Stopwatch: time between successive units leaving the step. Time 10+ consecutive cycles in 3 different periods; record min / avg / max.	
Shipping	Processing time (value-added) (min)	Time the product is actually being transformed at this step (mix time, bake time, freeze dwell). From recipe/spec and confirmed by observation.	
Shipping	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Shipping	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
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Shipping	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
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Shipping	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
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Shipping	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Shipping	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Shipping	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Shipping	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Shipping	Verify food-safety controls, traceability records	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Spaghetti diagram	Replace placeholder layout with plant drawing dimensions	Use the facility drawing; measure zone positions in feet.	
Spaghetti diagram	Trace walking, forklift, ingredient and product routes for one full shift	Follow each role with a layout printout; count trips; measure routes with a measuring wheel.	
Swimlane	Confirm the handoff and approval sequence with each department	Short interviews with Planning, Warehouse, Sanitation, Quality, Maintenance and the supervisors; mark duplicated work and waits.	
Food safety	Observe one allergen changeover and the sanitation verification	Record steps, times and sign-offs; compare with the allergen changeover procedure.	

Formulas

How each result is calculated

Calculation	Formula	Unit
Available time per day	Shifts/day × (Shift length " Planned breaks)	min/day
Takt time	Available time per day × 60 ÷ Customer demand per day	s/unit
Effective cycle time (per step)	Cycle time ÷ Uptime. Where uptime is not collected, cycle time is used and the result is marked Estimate.	s/unit
Bottleneck	Step with the highest effective cycle time among steps that apply to the line	
Line capacity	3600 ÷ Bottleneck effective cycle time	units/hr
Queue time (per step)	WIP before step ÷ Demand per day × Available time per day. If WIP was not counted, the observed Waiting time is used.	min
Total lead time	£ Queue time + £ Processing time over applicable steps	min
Value-added time	£ Processing time of steps classified Value-added	min
Non-value-added time	Total lead time " Value-added time	min
Process cycle efficiency (PCE)	Value-added time ÷ Total lead time	%
Rolled throughput yield	(Yield) over steps; where yield is missing, (1 " Scrap) is used	%
Changeover time per week	£ (Changeover time × Changeovers per week) ÷ 60	hr/week
Route distance	£ segment lengths of the drawn route × layout scale (1 unit = 1 ft)	ft/trip
Travel time per shift	Route distance × Trips per shift ÷ Speed for the movement category	min/shift
Annual travel labor hours	Travel time per shift × People × Shifts/day × Operating days/year ÷ 60	hr/yr
Annual travel labor cost	Annual travel labor hours × Loaded labor rate	$/yr
Estimated annual labor savings (travel)	Current annual travel labor cost " Future annual travel labor cost	$/yr
Changeover reduction	(Current " Future changeover hours/week) × Operating weeks/year; labor value = £ hours × step lab	hr/yr te
Scrap reduction opportunity	(Future RTY " Current RTY) × Demand/day × Operating days/year × Material cost/unit (valued at cos	$/yr
Throughput improvement potential	(Future capacity " Current capacity) ÷ Current capacity	%
Labor productivity	Line capacity ÷ Total labor on the line	units/labor-hr
PAM totals	£ activity time, distance and people-minutes by activity type (O, T, I, D, S); NVA time = £ time of a	min / ft ked N